

SIRAJUDEEN MUHAMMADU SHIMRIN

Embedded Systems & Hardware Engineering Undergraduate

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PROFESSIONAL SUMMARY

Computer Engineering undergraduate specializing in hardware design, embedded systems, and digital logic development. Experienced in HDL-based system design, FPGA concepts, and integrating hardware-oriented solutions with software technologies. Interested in developing efficient, reliable, and intelligent embedded systems.

EDUCATION

BSc Engineering (Computer Engineering)

University of Ruhuna, Sri Lanka

2023 – 2027

GPA: 3.43 / 4.00

Diploma in English Language and Literature

Aquinas College of Higher Studies, Sri Lanka

June 2024

Relevant Coursework: Digital Logic Design, Embedded Systems, Computer Architecture, Microprocessors, Computer Networks.

TECHNICAL COMPETENCIES

Hardware Design: Digital Logic Design, FPGA Concepts, Finite State Machines, RTL Design

HDL Programming: Verilog HDL, Hardware Description Languages, Simulation Concepts

Programming: C++, Python, Java, Bash/Shell

Tools & Systems: Git, GitHub, Linux, Simulation and Development Tools

HARDWARE ENGINEERING PROJECTS

HDL-Based 4-Floor Lift Controller | FPGA & Digital Design

- Designed a multi-floor elevator controller using Verilog HDL and Finite State Machine architecture.
- Implemented digital control logic and verified system behavior through simulation techniques.

Vision-Based Vehicle Speed Estimation | AI Embedded Application

- Developed an intelligent monitoring system combining hardware-oriented processing concepts with computer vision.
- Applied real-time image processing techniques for automated vehicle analysis.

Smart Nutrition Intelligence System (SNIS) | Intelligent Embedded Healthcare Concept

- Developed an intelligent healthcare-oriented system combining software and data-driven decision-making concepts.
- Explored the integration of computational techniques for real-world engineering applications.

Bike Race Network | Networked Application System

- Developed a communication-based platform demonstrating interaction between software systems and users.
- Applied system architecture concepts for designing connected applications.

Concrete Strength Prediction | Machine Learning Application

- Developed a predictive model demonstrating computational processing and engineering data analysis.
- Applied machine learning techniques for material property estimation.

EXTRACURRICULARS & COMMUNICATION

Vice-Captain, Karate Team | Faculty of Engineering, Ruhuna

2024 – 2025

Member | IESL & IEEE Student Branches

Active

Hardware Engineering Competencies:

FPGA Design

Verilog HDL

Digital Logic

Embedded Systems

Problem Solving

LANGUAGES

English (Professional), Sinhala (Native), Tamil (Native)